CC58 - Topics in Computer Science
Lab1: Graphics in gnuplot

Willy Ugarte

1 Install gnuplot

1.1 Download

For windows, go to https://sourceforge.net/projects/gnuplot/, download gnuplot install file and launch it.

2 Command Line

For windows, to configurate *gnuplot* as a command in your prompt requires to add it to your envirnement variables.

Thus, enter *sysdm.cpl* and add it to your system path.

3 Using gnuplot

3.1 Commands

In a terminal, run the gnuplot command and do the following exercises and questions:

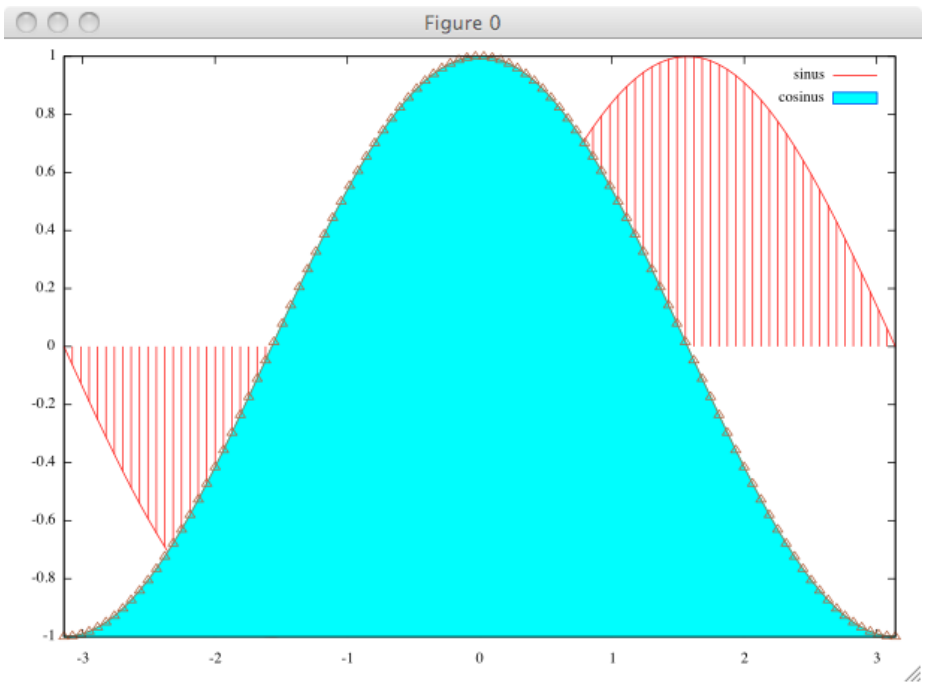
- `plot sin(x)`
- `plot sin(x) with lines`
- `plot sin(x) with impulses`
- `plot [-pi:pi] sin(x) with impulses`
- `plot [-pi:pi] sin(x) with impulses, sin(x) with lines`
- `plot [-pi:pi] sin(x) with impulses, sin(x) with lines, cos(x) with lines`
- `plot [-pi:pi] sin(x) with impulses, sin(x) with lines ls 1, cos(x) with lines`

- `plot [-pi:pi] sin(x) with impulses, sin(x) with lines ls 1, cos(x) with filledcurve fillstyle transparent pattern 4`
- `plot [-pi:pi] sin(x) with impulses, sin(x) with lines ls 1, cos(x) w filledc fs trans pat 4`
- `plot [-pi:pi] sin(x) with impulses notitle, sin(x) with lines ls 1 title \sinus", cos(x) w filledc fs trans pat 4 title \cosinus"`

3.2 Exercises

Exercise 1

Modify the last command so that the display obtained is exactly the same as the following. Use the documentation if necessary.



3.3 Work with data files

Exit gnuplot and restart it in order to start again in the initial configuration. Draw the sin (x) curve to verify that it works.

Download the following data files to your working directory:

- File1
- File2

- File3

Use the following command:

```
plot "fichier1.dat"
```

Is this the expected result? Read the documentation on the topic *using* in order to obtain a plot that displays the values of the 2nd column on the abscissa and those of the 3rd column column on the ordinate.

3.4 Scripts

From this demo-gnuplot.data.

The result is displayed in a window (Figure 1). By default, each data record is represented as a point and points are not linked. You can ask for a display that link points with lines (Figure 2), steps (Figure 3) or impulses (Figure 4) according to your needs. All these possibilities are described in the online help, obtained by the command help with.

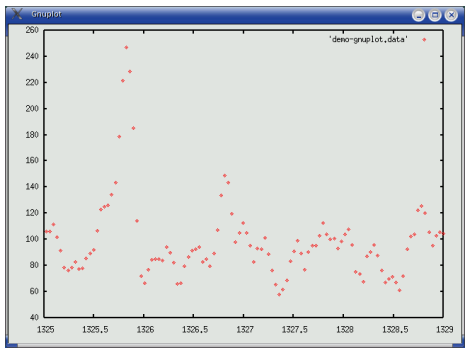


Figure 1: plot 'demo-gnuplot.data'

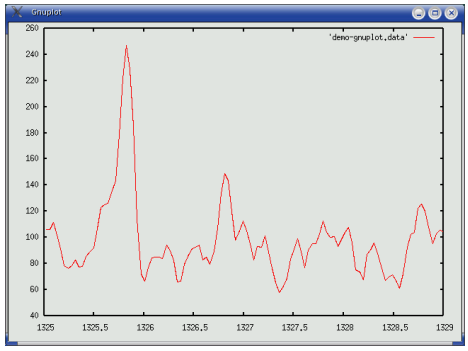


Figure 2: plot 'demo-gnuplot.data' with lines

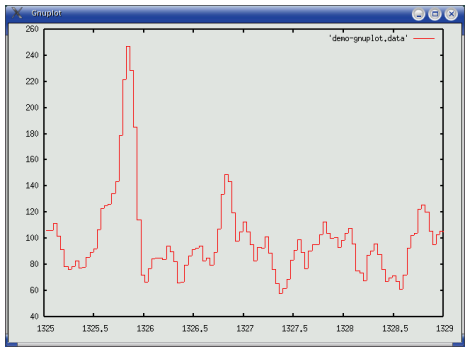


Figure 3: plot 'demo-gnuplot.data' with steps

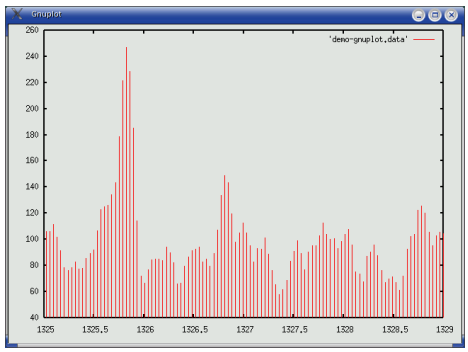


Figure 4: plot 'demo-gnuplot.data' with impulses

3.5 Coordinates of points

When your data file is composed of more than one column, you can choose to plot the second column according to the first one for instance. In that case each point drawn will use the first column as its x value and the second as its y value.

Use the data in data3.

```
plot 'data3' using 1:2 w l, 'data3' using 1:3 w l, 'data3' using
                        1:4 with lines
```

3.6 Changing the x and y ranges

It is sometimes necessary to modify the range of x and y values, for instance for zooming on a particular region or for the legend to be more visible. You can use the set xrange [interval] or set yrange for that:

```
set xrange[100:200] ; set yrange[0:900]
```

3.7 Labels and Titles

It is also possible to add some information on the graphics, we can for instance:

- add a grid for a better reading of the values: set grid;
- add a label on the x/y axes: set xlabel "poids moleculaire" set ylabel "volume"

```
plot 'demo-gnuplot.data' with lines
set grid
set xlabel "molecular weight"; set ylabel "volumes"
```

For adding titles:

```
set title "my title located top/centered"
```

You can also put some information directly within the graph for highlighting something by set label myLabel "the content of the label" at x,y justification (left/right/center). For removing the label just use the identifier of the label: set nolabel myLabel

```
set nogrid
set label 24 "< -- 247.024" at 1325.85,247 left
```

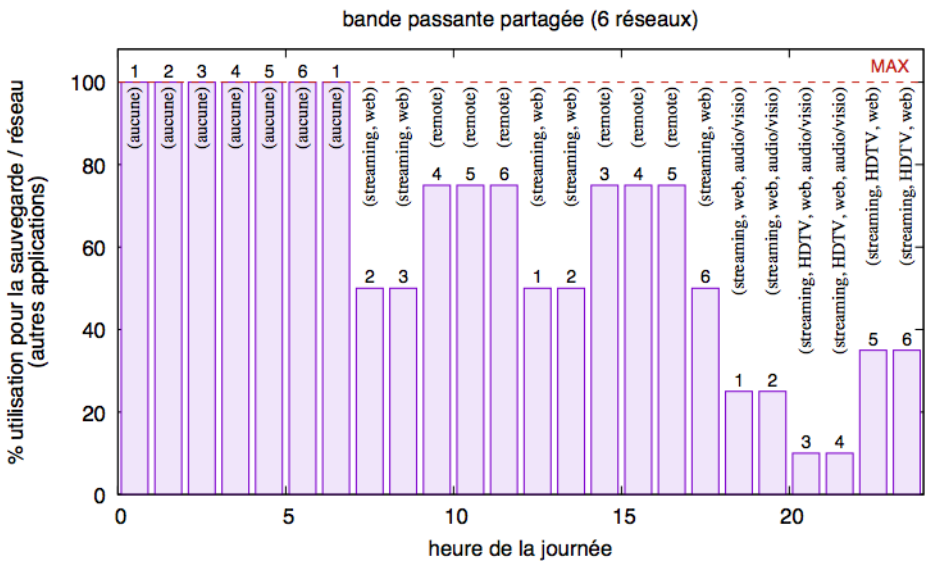
3.8 Producing images

Using gnuplot it is possible to produce images directly. For that it is necessary to inform gnuplot of the type of images we want to produce by providing it the output type (png for instance) and output file (image.png for instance).

```
set grid
set xlabel "molecular weigth"
set ylabel "volumes"
set terminal png
set output 'demo2.png'
plot 'demo-gnuplot.data2' with impulses
```

Exercise 2

From this data. Reproduce the following plot (by choosing to save the image in PDF format):



Exercise 3

Download the data from the blackboard of the students in the computer science program, and make a plot representing the distribution of male/female students.

Exercise 4

Download the data from `http://sigmed.minedu.gob.pe/descargas/`.

- Make a plot of all the schools in Peru.
- Make a plot of all the schools in Lima.
- Make a plot of all the schools in your neighborhood.